Corporate Finance

Lecture 4 extras: Read the news, and look around you

Camilo Saldarriaga

Finance Department, ESCP Business School, 2026/2027

The Big Short (2015): a true story



- Based on the book by Michael Lewis. A handful of investors bet **against** the US housing market in 2005–2007, and won when it collapsed.
- The whole film turns on two strange names: **subprime mortgage-backed securities** and **CDOs**.
- Behind them sits an idea you already know from Lectures 2 and 3: **correlation** and portfolio risk.

Subprime MBS and CDOs, in plain words

- **Mortgage**: a loan a family takes to buy a house, repaid every month. **Subprime**: a mortgage given to a borrower with a weak credit history, who is more likely to stop paying.
- **Mortgage-backed security (MBS)**: a bank bundles thousands of mortgages into one pool and sells pieces of it to investors. The investors receive the monthly repayments of the families.
- **Collateralized debt obligation (CDO)**: a bundle of those bundles, cut into layers (*tranches*). When families stop paying, the losses hit the bottom layer first. The top layer is paid first and loses money only if **many** mortgages fail **at the same time**.
- Rating agencies gave that top layer a **AAA** rating: as safe as US government bonds. The question is: why?

Why AAA? Correlation, again

- Recall the variance of a portfolio: the comovement term, with the correlation ρ, decides how much risk diversification removes.

Normal times: low correlation. One family loses its job in Ohio, another gets divorced in Florida. Defaults are mostly **unrelated**, so in a pool of thousands a few fail and most pay. The top layer almost never loses: AAA.

Crisis: high correlation. House prices fall **everywhere at once**. Families cannot sell or refinance, and they default **together**. The correlation jumps towards 1, diversification disappears, and even the "safe" top layers lose money.

- The models used the correlation of good times, so the real risk was **much greater than estimated**. Correlation rises exactly when you need diversification most.